Summer Math Adventure

Weeks 6 – 8 Extension Booklet

Exponents · Expressions & Equations · Probability

3 Weeks · 15 Daily Pages · 3 Quizzes · Full Answer Key · ~45 min/day

Name: ________________________________

Summer: 2025

☐ Week 6: Exponents	☐ Week 6 Quiz
☐ Week 7: Expressions & Eq.	☐ Week 7 Quiz
☐ Week 8: Probability	☐ Week 8 Quiz

Tools needed: pencil, calculator (Week 6 challenge only)

Summer Math Adventure

W6
D1

Week 6 · Day 1 | What Is an Exponent?

Name: ____________________ Date: ____________________

EXPONENTS

Key Idea: An *exponent* tells you how many times to multiply the *base* by itself.

3^4 means 3 × 3 × 3 × 3 = **81** | base = 3, exponent = 4

Special names: 5^2 = "five squared" | 5^3 = "five cubed"

Remember: Any number to the power of 0 equals 1. (7^0 = 1)

Part A – Label the Parts

For each expression, write the base, the exponent, and read it aloud.

1. 2^5

Base: ____________________

Exponent: ____________________

Read as: ____________________

2. 10^3

Base: ____________________

Exponent: ____________________

Read as: ____________________

3. 6^2

Base: ____________________

Exponent: ____________________

Read as: ____________________

4. 4^0

Base: ____________________

Exponent: ____________________

Value: ____________________

Part B – Expand & Evaluate

Write as repeated multiplication, then find the value.

5. 3^3 = ___ × ___ × ___ =

____

6. 2^6 = ___ × ___ × ___ × ___ × ___ × ___ =

____

7. 5^2 = ___ × ___ =

8. 10^4 = ___ × ___ × ___ × ___ =

Summer Math Adventure

W6
D2

Week 6 · Day 2 | Powers of 10

Name: ______________________ Date: ______________________

EXPONENTS

Pattern: $10^1=10$ | $10^2=100$ | $10^3=1{,}000$ | $10^4=10{,}000$

Each time the exponent goes up by 1, you multiply by 10 (add a zero).

Shortcut: 10^n = 1 followed by *n* zeros.

Part A – Complete the Powers of 10 Table

Exponential	Expanded	Standard Form	Number of Zeros
10^1	10	10	1
10^2	10×10		
10^3			
10^4			
10^5			
10^6			

Part B – Multiply Using Powers of 10

Use the pattern: $4 \times 10^3 = 4 \times 1{,}000 = 4{,}000$

1. $7 \times 10^2 =$ ______________________

2. $3 \times 10^4 =$ ______________________

3. $9 \times 10^1 =$ ______________________

4. $6 \times 10^5 =$ ______________________

5. $25 \times 10^3 =$ ______________________

6. $14 \times 10^2 =$ ______________________

Part C – Fill in the Exponent

7. $10^{\boxed{}} = 1{,}000{,}000$

8. $10^{\boxed{}} = 100{,}000$

Summer Math Adventure

W6 D3

Week 6 · Day 3 | Comparing Exponential Expressions

Name: ______________________ Date: ______________________

EXPONENTS

Strategy: Always evaluate each expression first, then compare.

Example: Is 2^5 or 5^2 greater?

$2^5 = 32$ $5^2 = 25$ So $2^5 > 5^2$

Part A – Evaluate Each Side, Then Compare (<, >, =)

1. $3^4 \;\square\; 4^3$

3^4 = ___ 4^3 = ___

2. $2^7 \;\square\; 7^2$

2^7 = ___ 7^2 = ___

3. $5^3 \;\square\; 3^5$

5^3 = ___ 3^5 = ___

4. $6^2 \;\square\; 2^6$

6^2 = ___ 2^6 = ___

5. $10^3 \;\square\; 1{,}000$

6. $4^2 \;\square\; 2^4$

Part B – Order from Least to Greatest

7. 2^3 3^2 2^4 4^1
Values: ___, ___, ___, ___
Order:

____

8. 5^0 2^5 3^3 10^2
Values: ___, ___, ___, ___
Order:

____

Part C – True or False? Correct the False Ones.

9. $2^3 = 6$ T / F
If false, correct:

____

10. $3^0 = 0$ T / F
If false, correct:

____

11. $10^3 = 1{,}000$ T / F

12. $1^{100} = 100$ T / F

Summer Math Adventure

W6 D4

Week 6 · Day 4 | Order of Operations with Exponents

Name: ____________________ Date: ____________________

EXPONENTS

PEMDAS Order:

1. **P**arentheses 2. **E**xponents 3. **M**ultiplication / **D**ivision (left→right) 4. **A**ddition / **S**ubtraction (left→right)

Example: $2 + 3^2 \times 4 = 2 + 9 \times 4 = 2 + 36 =$ **38**

Part A – Evaluate Step by Step

Show each step. Circle the operation you do first.

1. $5 + 2^3$
Step 1: ___
Answer:

____

2. $4^2 - 10$
Step 1: ___
Answer:

____

3. 3×2^4
Step 1: ___
Answer:

____

4. $(3 + 1)^2$
Step 1: ___
Answer:

____

5. $2^3 + 3^2$
Step 1: ___ Step 2: ___
Answer:

____

6. $50 - 2^2 \times 3$
Step 1: ___ Step 2: ___ Step 3: ___
Answer:

____

Part B – Multi-Step Expressions

7. $(2 + 3)^2 - 4 \times 3 =$

____

8. $6^2 \div (3 + 1) + 5 =$

____

Summer Math Adventure

W6 D5

Week 6 · Day 5 | Perfect Squares & Square Roots

Name: ______________________ Date: ______________________

EXPONENTS

Perfect Square: A number that equals an integer times itself. $25 = 5 \times 5 = 5^2$

Square Root (√): The opposite — what number times itself gives you this? $\sqrt{25} = 5$

First 12 perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144

Part A – Perfect Squares Chart

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2	1											

Part B – Square Roots

1. $\sqrt{36}$ = ______________________
2. $\sqrt{81}$ = ______________________
3. $\sqrt{144}$ = ______________________
4. $\sqrt{49}$ = ______________________
5. $\sqrt{121}$ = ______________________
6. $\sqrt{1}$ = ______________________

Part C – Is It a Perfect Square?

Circle YES or NO. If yes, write its square root.

7. 50 YES / NO √ = ___
8. 64 YES / NO √ = ___
9. 100 YES / NO √ = ___
10. 72 YES / NO √ = ___

Part D – Area & Square Roots in Context

11. A square garden has an area of 64 m². What is the length of one side?
Side = ______________________

12. A checkerboard has 64 small squares arranged in a perfect square pattern. How many rows does it have?
Rows = ______________________

Summer Math Adventure

W6 QUIZ

Week 6 Quiz | Exponents

Name: ______________________ Date: ______________________

Score: ______ / 20 Time started: _______ Finished: _______

Section 1 – Evaluate (1 pt each)

1. 4^3 =

2. 2^7 =

3. 9^0 =

4. 10^5 =

5. $\sqrt{64}$ =

6. $\sqrt{121}$ =

Section 2 – Order of Operations (2 pts each)

7. $2^4 + 6 \times 3$ =

8. $(5 - 2)^3 \div 9$ =

9. $5^2 - 3^2 + 4$ =

10. $100 \div 10^2 \times 3$ =

Section 3 – Compare & Reason (2 pts each)

11. Circle the greater value: 3^5 or 5^3 Show your work below.

12. A square room has an area of 196 sq ft. What is the length of each wall? Show work.

Summer Math Adventure

W7
D1

Week 7 · Day 1 | Variables & Algebraic Expressions

Name: ____________________ Date: ____________________

EXPRESSIONS

Variable: A letter that stands for an unknown number. (x, n, a, b...)

Expression: Numbers, variables, and operations — but NO equals sign. (3x + 5)

Term: Parts of an expression separated by + or –. In 3x + 5: terms are 3x and 5

Coefficient: The number multiplied by a variable. In 3x, the coefficient is 3.

Part A – Vocabulary Match

Label each part of the expression: 4n – 7 + 2n

Expression: **4n – 7 + 2n**

Number of terms:

____

List all terms:

____

Coefficient of the first term:

____

Constant (number with no variable):

____

Part B – Write as an Algebraic Expression

Let n = the unknown number.

1. A number plus 8 →

____

2. 5 times a number →

____

3. A number divided by 4 →

____

4. 12 minus a number →

____

5. Twice a number, plus 3 →

6. A number squared, minus 1 →

Summer Math Adventure

W7 D2

Week 7 · Day 2 | Simplifying Expressions – Like Terms

Name: ____________________ Date: ____________________

EXPRESSIONS

Like Terms have the same variable (and exponent). You can combine them by adding/subtracting coefficients.

$3x + 5x = 8x$ | $7a - 2a = 5a$ | *BUT:* $3x + 5y$ cannot be combined — different variables!

Tip: Constants (plain numbers) are also like terms with each other: $4 + 9 = 13$

Part A – Identify Like Terms

Group the like terms by circling them with matching colors (or labels A, B, C...).

1. $3x$ $5y$ $2x$ 8 $4y$ 11 $7x$

Like-term groups:

____

2. $6a$ $2b$ $3a^2$ $4a$ $9b$ a^2

Like-term groups:

____

Part B – Combine Like Terms

3. $4x + 3x =$ ____________________

4. $9y - 2y =$ ____________________

5. $5a + 2a + 3a =$ ____________________

6. $8m - 5m + m =$ ____________________

7. $3x + 4 + 2x + 7 =$ ____________________

8. $6n - 2 + n + 5 =$ ____________________

9. $4a + 3b - a + 2b =$ ____________________

10. $10x + 5 - 4x - 3 =$ ____________________

Part C – Simplify, Then Evaluate

Summer Math Adventure

W7 D3

Week 7 · Day 3 | Writing & Solving One-Step Equations

Name: ____________________ Date: ____________________

EQUATIONS

Equation = two expressions set equal. There IS an = sign. $3x + 5 = 20$

Solve = find the value of the variable that makes it true.

Strategy (inverse operations): Whatever is done to the variable, do the opposite to both sides.

$x + 6 = 14$ → subtract 6 from both sides → $x = 8$

$3x = 21$ → divide both sides by 3 → $x = 7$

Part A – Name the Inverse Operation

1. $x + 9 = 15$ → use: ____________________

2. $x - 4 = 11$ → use: ____________________

3. $5x = 35$ → use: ____________________

4. $x \div 3 = 8$ → use: ____________________

Part B – Solve. Show Your Check.

5. $x + 12 = 30$
$x =$ ____________________

Check: ___ *+ 12 = 30 ✓*

6. $y - 7 = 19$
$y =$ ____________________

Check: ___ *− 7 = 19 ✓*

7. $4n = 48$
$n =$ ____________________

Check: ___

8. $\frac{m}{6} = 9$
$m =$ ____________________

Check: ___

9. $t + 25 = 100$
$t =$ ____________________

10. $8p = 72$
$p =$ ____________________

Summer Math Adventure

W7 D4

Week 7 · Day 4 | Two-Step Equations

Name: ______________________ Date: ______________________

EQUATIONS

Two steps: undo addition/subtraction FIRST, then multiplication/division.

Example: $2x + 3 = 11$

Step 1: Subtract 3 from both sides → $2x = 8$

Step 2: Divide both sides by 2 → $x = 4$

Always check: $2(4) + 3 = 8 + 3 = 11$ ✓

Part A – Label the Steps

Solve $3n - 5 = 16$. Fill in the blanks:
Step 1: Add ___ to both sides → 3n = ___
Step 2: Divide both sides by ___ → n = ___
Check: 3(___) – 5 = ___ ✓

Part B – Solve. Show All Steps & Check.

1. $2x + 7 = 21$

x = ______________________

2. $4y - 3 = 17$

y = ______________________

3. $5m + 10 = 45$

m = ______________________

4. $\frac{n}{3} + 4 = 10$

n = ______________________

5. $6a - 12 = 0$

6. $3t + 15 = 60$

Summer Math Adventure

W7
D5

Week 7 · Day 5 | Inequalities

Name: ______________________ Date: ______________________

EQUATIONS

Inequality uses <, >, ≤, ≥ instead of =. The answer is a range of values.

$x + 3 > 7 \rightarrow x > 4$ (any number greater than 4 works)

Graphing: Open circle ○ for < / > (not equal) · Closed circle ● for ≤ / ≥ (includes the value)

Arrow points in the direction of solutions.

Part A – Write the Inequality

1. "x is greater than 5" →

____

2. "y is at most 12" →

____

3. "n is less than or equal to 8" →

____

4. "t is at least 20" →

____

Part B – Solve the Inequality

5. $x + 4 > 9$ →

____

6. $y - 3 \leq 7$ →

____

7. $2n < 16$ →

____

8. $\frac{m}{4} \geq 3$ →

____

9. $3x - 1 > 11$ →

____

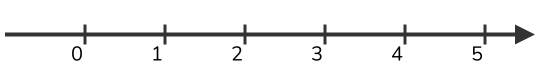
10. $5 + 2y \leq 19$ →

____

Part C – Graph the Solution on the Number Line

Draw an open or closed circle, then shade the arrow.

11. $x > 3$



12. $y \leq 2$

Summer Math Adventure

W7 QUIZ

Week 7 Quiz | Expressions & Equations

Name: ______________________ Date: ______________________

Score: ______ / 20 Time started: _______ Finished: _______

Section 1 – Expressions (1 pt each)

1. Write an expression: "6 less than 3 times x"

____

2. Evaluate $4n - 2$ when $n = 7$ =

____

3. Simplify: $5x + 3 + 2x - 1 =$

____

4. Simplify: $6a - 2b + a + 3b =$

____

Section 2 – Equations (2 pts each)

5. $x + 17 = 40$

$x =$ ______________________

6. $7y = 63$

$y =$ ______________________

7. $3m + 8 = 29$

$m =$ ______________________

8. $\frac{n}{4} - 2 = 5$

$n =$ ______________________

Section 3 – Word Problem & Inequality (2 pts each)

9. A box of crayons costs $5. Lena also buys a notebook for $3. She spends $23 total. How many boxes of crayons did she buy?

Equation:

____

Summer Math Adventure

W8
D1

Week 8 · Day 1 | What Is Probability?

Name: ______________________ Date: ______________________

PROBABILITY

Probability = how likely an event is to happen.

P(event) = $\dfrac{\textbf{number of favorable outcomes}}{\textbf{total number of possible outcomes}}$

Scale: 0 (impossible) ←——————————→ 1 (certain)

P = 0.5 or $\dfrac{1}{2}$ means equally likely as unlikely.

Part A – Probability Scale

Place each event on the probability scale below. Write the letter at the right spot.

mpossible — Certain

0 — 1/2 — 1

A. Rolling a 7 on a standard die

B. Flipping heads on a fair coin

C. The sun rises tomorrow

D. Picking a red card from a standard deck

E. It will snow in Arizona in July

F. Rolling an even number on a die

Part B – Calculate the Probability

A bag has 3 red, 4 blue, and 5 green marbles. Write each probability as a fraction.

1. P(red) = ______________________

2. P(blue) = ______________________

3. P(green) = ______________________

4. P(NOT red) = ______________________

5. P(red or blue) = ______________________

6. P(purple) = ______________________

Part C – Probability as Decimal and Percent

Use the same bag (3 red, 4 blue, 5 green marbles). Express each probability as a fraction, decimal, and percent.

7. P(green):

Fraction:

8. P(blue):

Fraction:

Summer Math Adventure

W8 D2

Week 8 · Day 2 | Experimental vs. Theoretical Probability

Name: ____________________ Date: ____________________

PROBABILITY

Theoretical Probability: What SHOULD happen based on math. P(heads) = $\frac{1}{2}$

Experimental Probability: What ACTUALLY happens when you do the experiment.

If you flip a coin 20 times and get 12 heads → P(heads) = $\frac{12}{20} = \frac{3}{5}$

The more trials you run, the closer experimental gets to theoretical.

Part A – Coin Flip Experiment Data

Omar flipped a coin 30 times. Results are below.

Outcome	Tally	Frequency
Heads	|||| |||| |||| |	
Tails	|||| |||| |||| ||||	

1. Experimental P(Heads) = ____________________

2. Experimental P(Tails) = ____________________

3. Theoretical P(Heads) = ____________________

4. Are they the same? Why/why not? ____________________

Part B – Die Roll Experiment

A die was rolled 60 times. Fill in the theoretical, then compare.

Number	1	2	3	4	5	6
Experimental Freq.	12	8	11	9	13	7
Experimental Prob.						
Theoretical Prob.						

5. Which number had experimental probability closest to theoretical?

____

6. If you rolled 600 times instead of 60, would you expect experimental to be closer to or farther from theoretical? Explain.

Summer Math Adventure

W8
D3

Week 8 · Day 3 | Sample Space & Tree Diagrams

Name: ____________________ Date: ____________________

PROBABILITY

Sample Space: All possible outcomes of an experiment.

Tree Diagram: A visual tool to list all outcomes systematically.

Example – Flip a coin AND roll a die:

H → 1, 2, 3, 4, 5, 6 T → 1, 2, 3, 4, 5, 6 Total: 12 outcomes

Counting Principle: # of outcomes = options × options × ...

Part A – List the Sample Space

1. Flip a coin (H/T) and spin a spinner with colors Red, Blue, Green.
Complete the sample space:
{ ___, ___, ___, ___, ___, ___ }
Total outcomes: ___

Part B – Draw a Tree Diagram

2. A lunch special has 2 sandwich choices (Turkey, Veggie) and 3 drink choices (Water, Juice, Milk). Draw a tree diagram and list ALL combinations.

Total combinations: ___

Part C – Use the Counting Principle

3. A shirt comes in 4 colors and 3 sizes. How many combinations?

____

4. You roll 2 dice. How many total outcomes?

____

5. A password uses 1 letter (A–E) and 1 digit (1–5). How many passwords?

____

6. You flip 3 coins. How many total outcomes?

____

Part D – Probability from a Sample Space

Use the Turkey/Veggie × Water/Juice/Milk sample space from Part B.

7. P(Turkey AND Juice) =

8. P(any Veggie combo) =

Summer Math Adventure

W8 D4

Week 8 · Day 4 | Making Predictions with Probability

Name: ______________________ Date: ______________________

PROBABILITY

Prediction Formula: Expected outcomes = P(event) × number of trials

Example: P(rolling a 3) = $\frac{1}{6}$. If you roll 60 times: expected 3s = $\frac{1}{6}$ × 60 = **10**

This is a *prediction*, not a guarantee — actual results may vary!

Part A – Calculate Expected Outcomes

A spinner has 5 equal sections: 1 Red, 1 Blue, 2 Yellow, 1 Green.

1. P(Red) = ______________________

2. P(Yellow) = ______________________

3. If you spin 50 times, expected # of Red: ______________________

4. If you spin 50 times, expected # of Yellow: ______________________

5. If you spin 100 times, expected # of Green: ______________________

6. If you spin 30 times, expected # of Blue: ______________________

Part B – Real-World Predictions

7. A factory knows 3 out of every 200 light bulbs are defective. If they produce 1,000 bulbs, how many do they expect to be defective?

Answer: ______________________

8. A survey of 50 students showed 20 prefer math class. If there are 350 students in the school, how many would you predict prefer math?

Answer:

Summer Math Adventure

W8
D5

Week 8 · Day 5 | Probability Review & Mixed Practice

Name: ____________________ Date: ____________________

PROBABILITY

Week 8 Summary:

✓ P(event) = favorable ÷ total | ✓ 0 = impossible, 1 = certain

✓ Theoretical vs. Experimental | ✓ Tree diagrams = list all outcomes

✓ Predictions = P × trials | ✓ Complement: P(NOT event) = 1 – P(event)

Part A – Mixed Probability Problems

A standard deck has 52 cards: 4 suits (♠ ♥ ♦ ♣), 13 cards each, including Ace, 2–10, Jack, Queen, King.

1. P(drawing a heart) = ____________________

2. P(drawing a King) = ____________________

3. P(drawing a red card) = ____________________

4. P(NOT a spade) = ____________________

5. P(a Jack or Queen) = ____________________

6. P(a number card 2–9 of clubs) = ____________________

Part B – Complement Rule

P(NOT event) = 1 – P(event)

7. P(rain) = 0.3. P(no rain) = ____________________

8. P(win) = $\frac{2}{5}$. P(not win) = ____________________

9. P(red marble) = $\frac{3}{8}$. P(not red) = ____________________

10. P(NOT rolling a 6) = ____________________

Part C – Scenario & Decision

11. A raffle sells 200 tickets. You buy 5. What is the probability you win?

P(win) =

____

Summer Math Adventure

W8 QUIZ

Week 8 Quiz | Probability

Name: ______________________ Date: ______________________

Score: ______ / 20 Time started: _______ Finished: _______

Section 1 – Probability Calculations (1 pt each)

A jar has 6 red, 3 blue, and 1 yellow marble (10 total).

1. P(red) = ______________________
2. P(blue) = ______________________
3. P(yellow) = ______________________
4. P(NOT red) = ______________________
5. P(red or blue) = ______________________
6. P(green) = ______________________

Section 2 – Tree Diagram & Counting (2 pts each)

7. You choose a flavor (Chocolate, Vanilla) and a cone (Waffle, Sugar, Cup). Draw a tree diagram and list all combinations.

Total: ___

8. From that same sample space, what is P(Chocolate AND Waffle cone)?
P = ______________________

Section 3 – Prediction & Reasoning (2 pts each)

9. P(landing on blue) = $\frac{1}{4}$. If you spin 80 times, how many blues do you expect?

Answer:

____

10. A survey found 35 out of 100 students walk to school. Predict how many of 400 students walk

Answer Key – Weeks 6, 7 & 8

For parent/teacher use only. Cut or fold before giving booklet to student.

WEEK 6 – EXPONENTS

Day 1 – What Is an Exponent?

Part A:
1. base=2, exp=5, "two to the fifth power"
2. base=10, exp=3, "ten cubed"
3. base=6, exp=2, "six squared"
4. base=4, exp=0, value=1

Part B:
5. 3×3×3=27 6. 64 7. 25
8. 10,000 9. 1 10. 64

Part C:
11. 7^3 12. 9^2 13. 2^5 14. 10^3

Challenge: $2^8=256$, $4^4=256$ — equal!

Day 2 – Powers of 10

Table: 100; 10×10×10=1,000; 10,000; 100,000; 1,000,000

Part B:
1. 700 2. 30,000 3. 90
4. 600,000 5. 25,000 6. 1,400

Part C: 7. 6 8. 5 9. 4 10. 2
Part D: 11. 300 12. 200,000 13. 300,000,000

Day 3 – Comparing Expressions

Part A:
1. 81 > 64 so > 2. 128 > 49 so >
3. 125 < 243 so < 4. 36 < 64 so <
5. = 6. =

Part B:
7. 8, 9, 16, 4 → 4, 8, 9, 16
8. 1, 32, 27, 100 → 1, 27, 32, 100

Part C:
9. F (=8) 10. F (=1) 11. T 12. F (=1)

Day 4 – PEMDAS with Exponents

Part A:
1. 5+8=13 2. 16−10=6 3. 3×16=48
4. $(4)^2=16$ 5. 8+9=17 6. 50−12=38

Part B:
7. 25−12=13 8. 36÷4+5=14
9. 27−18=9

Part C:
10. $(3+5)^2\times2=128$
11. $4\times(2+1)^2=36\neq9$; try $4\times2+1=9$ ✓

Day 5 – Perfect Squares & Square Roots

Chart: 1,4,9,16,25,36,49,64,81,100,121,144

Part B: 1.6 2.9 3.12 4.7 5.11 6.1
Part C: 7.No 8.Yes,8 9.Yes,10 10.No

Part D: 11. 8m 12. 8 rows 13. 81 cm²
Challenge: $7^2=49$, $8^2=64$, so $7<\sqrt{50}<8$

Week 6 Quiz

Sec 1: 1.64 2.128 3.1 4.100,000 5.8 6.11

Sec 2: 7. 16+18=34 8. 27÷9=3 9. 25−9+4=20 10. 100÷100×3=3

Sec 3: 11. $3^5=243 > 5^3=125$; circle 3^5 12. $\sqrt{196}=14$; each wall = 14 ft

WEEK 7 – EXPRESSIONS & EQUATIONS